TAVI: UB and A1-A6 migration audit

Technical closeout review · 12 October 2026

Sign-off remains open. The core UB, stage and angle mathematics passed independent numerical checks, and real McStas controls support the nominal and tilted scattering paths. The programme still has three introduced functional or verification gaps, four inherited functional defects, a documented scientific limitation, and an unfinished downstream migration. These findings do not justify rolling back the mathematical changes; they do justify a focused closeout before declaring the work complete.

Scope and decision

This audit pins TAVI main at **97a295b3d6167843b0f4259aa79b284dc55a3857**. It compares the state before the UB work, the completed UB slices, the state before the angle programme, and the current implementation. It covers the four shipped instruments: PUMA, IN8, IN12 and PANDA.

The review combined source and history inspection, a complete repository test run, independent geometry and fitting calculations, real offscreen GUI/controller reproductions, current ISAR consumer inspection, and serial McStas 3.4 compilation and execution. Production repository files were left unchanged. The evidence bundle contains the scripts, numerical outputs, test logs and selected generated-model outputs supporting the conclusions.

The recommended immediate repair order is: silent state/data loss, the three introduced gaps, the remaining scan precision problem, and the consumer migration. The analytic-frame approximation requires an explicit scientific scope decision. It should remain visible wherever general UB mounts are used for quantitative resolution-convolved predictions.

Findings at a glance

ID	Finding	Origin	Recommended disposition
F1	PATCH can report success while later callbacks overwrite explicit requested values	Inherited	High closeout priority: remote state must match its acknowledgement
F2	Saving can lose recorded or edited UB observations	Inherited	High closeout priority: preserve alignment work before fitting
F3	ISAR client/parser/grid contract is not migrated to current TAVI	Incomplete planned U5	Completion blocker for the integrated workflow
F4	Direct A2/A6 edits leave public Q and HKL stale	Inherited	Fix coordinate/readout synchronization
F5	Explicit zero A2/A6 API launch is rejected despite supported transmission semantics	New in angle U3	Repair supported launch path
F6	Manual tilted-reflection gate does not install its described physical truth	New during UB truth separation	Repair acceptance fixture before using it for sign-off
F7	Negative slit scan gaps are accepted and emitted	New scan surface in U3	Validate physical gap domain after relative expansion
F8	Fine scans silently collapse to repeated positions	Inherited	Preserve calculation precision or explicitly reject unsupported steps
F9	Analytic convolution still treats H/K/L as instrument resolution directions	Known retained approximation	Scientific limitation: implement frame mapping or retain an explicit restricted scope

The priority labels above express closeout order, not a claim that all findings were introduced by the recent commits. F9 is already documented. F3 is an anticipated compatibility window whose final phase has not landed. F1, F2, F4 and F8 exist in the pre-UB baseline.

1. What changed, and what is mathematically sound

Historical boundaries

Boundary	Pinned revision	Meaning
Before UB	971a91e96cf7e3ba734d3287c75ef048248b66ce	Baseline used to distinguish inherited behavior
UB stage core	8b04f18499b4da66572c73ebce52ab76e7b9508d	PR 48, merged 1 October
UB truth and locking	e4064c9c7798cec95110c2d76035ce245842dce1	PR 49, merged 1 October
UB residual/refinement closeout	010cec6d738871830cac44796b1b0cd591fe81a0	PR 50, merged 1 October
Before angle programme	4547acb69d0117839ba7d2b96e20fc43ab594c3d	Baseline before U1
Before angle U3	869ad3b6958e03ee185db061025634387c583acf	U2 complete; direct comparison for F5
Audited main	97a295b3d6167843b0f4259aa79b284dc55a3857	U1-U4 merged, plus closeout documentation

The angle programme's PRs 55-58 landed on 9-10 October. The latest inspected release is v1.3.2, published 7 October, at commit 26c43f49116a992b7d335378a8590aa8032e0df8. Thus the inspected release contains the UB work but predates the angle programme. A successful audit of main should not be described as proof that the angle changes are already in that release.

Current angle contract

Angle	Canonical quantity	Physical meaning
A1	mono_theta_deg	Monochromator Bragg angle; derived as A2/2
A2	mono_two_theta_deg	Monochromator take-off, 2 theta
A3	sample_rotation_deg	Sample turntable rotation about the vertical
A4	sample_two_theta_deg	Sample scattering angle, 2 theta
A5	analyzer_theta_deg	Analyzer Bragg angle; derived as A6/2
A6	analyzer_two_theta_deg	Analyzer take-off, 2 theta

The former internal/public four-angle convention called monochromator take-off A1, sample two-theta A2, sample rotation A3, and analyzer take-off A4. The current mapping is therefore a semantic migration, not a cosmetic relabeling. A1 and A5 are now dependent readouts, and refusing them as independent scan inputs is intentional. The lower and upper sample arcs, sgl and sgu, have no A number. S01

The inspected names agree across the registry, API conversion, scan planner and generated McStas parameters. Aliases such as mtt, sth, stt and att name the same physical quantities; they do not implement a facility encoder sign or zero transformation. The signs remain descriptor-owned: PUMA (+,-,+), IN8 (+,+,-), IN12 (-,+,-), PANDA (-,+,-) for A2/A4/A6. Retiring correction-angle meanings and refusing old state/output schemas are approved boundaries, not regressions. S01 S19

Geometry and truth

The core uses Q_mount = U B hkl, with the reciprocal factor 2 pi included in B. Lab/mount coordinates put y vertical and z along the incident beam. The public Q interface swaps the last two components so its z is vertical. That swap is a coordinate convention; treating it as a proper rotation would introduce a false audit finding. S01

The stage composition is R_stage = R_A3(y) R_sgl(x) R_sgu(z), outermost first. The physical crystal is U_true = R_hidden U_described. The operator's fitted UB remains a belief used for targeting and inverse interpretation. The emitted McStas local-frame transform is (R_stage U_true)^T. Independent matrix reconstruction confirmed this transpose and rotation order. S24

The UB work made real corrections: hidden training orientation no longer directly becomes the operator's UB; the physical arm reads the true mount; and both peak inverse paths consistently undo the positive-sense Friedel sign. Synthetic fits on both sample senses recovered the intended U, including nonorthogonal cells. Removing hidden motor-zero errors from the mount-only exercise is coherent: a fixed mount rotation can be represented by one U, whereas a general motor-zero error across tilted configurations cannot. S01 S24 S25

I found no new failure in reciprocal metrics, the shipped crossed-arc reachability geometry, orientation fitting, lattice refinement, plane locking, or the physical center transform. The scope of that positive conclusion is the tested geometry and shipped-stage behavior, not every possible future sample component.

2. Confirmed functional and integration findings

F1. A successful PATCH can silently discard explicit requested values

Origin: inherited. Priority: repair before relying on remote state acknowledgements.

The code documents a dependency order: sample/lattice, energy mode, Q/HKL, then angles. However, it writes all requested values first and only then fires the collected callbacks. An earlier callback can recompute a field explicitly set later in that order. The acknowledgement retains the requested value in its applied list even though the actual state has changed. S02 S03

Actual offscreen IN8 controller results:

Submitted values	Reported result	Actual state afterward
lattice_a_angstrom=4.15, h=1	Both applied; no errors	h=2.0494
energy_transfer_mev=1, A4=40	Both applied; no errors	sample_two_theta_deg=69.8827
h=1, A3=35	Both applied; no errors	sample_rotation_deg=16.9322
h=1, sample_lower_arc_deg=2	Both applied; no errors	sample_lower_arc_deg=0

The lattice-plus-HKL example is an ordinary, meaningful update and is sufficient to establish the defect independently of any policy about conflicting coordinate systems. Single A4 updates and a combined A3/A4 update worked as controls.

Required behavior: apply coordinated dependency groups without allowing their derived updates to erase later explicit inputs. Define and enforce precedence or refusal for contradictory coordinate requests. Verify the resulting public state, rather than only the applied list. Do not reuse the launch builder unchanged: it begins from defaults, while PATCH must begin from current state.

Acceptance check: submit a lattice-plus-HKL update, then read parameters back and independently solve the resulting geometry. Repeat with the supported mixed groups. The acknowledgement and final state must agree within declared numerical/display tolerances.

Evidence: interfaces/repro_interfaces.py and its BATCH_PATCH log records. The all-setters-then-all-callbacks mechanism is already present at the pre-UB baseline.

F2. Save does not preserve the current UB observation list

Origin: inherited. Priority: repair before alignment work is treated as safely saved.

Take Position and subsequent edits update the dock's observations. The UB model's peak list is copied from the dock only during Calculate UB or Refine Lattice. Save serializes the model list, which can still be empty or reflect an older fit. S04 S05

Reproduction: start IN8, set the first peak to (2,0,0), take its position, lock it, save, and restart. The dock initially has valid stage angles, ki=kf=2.6634, and a locked observation. The saved ub_matrix_state.peaks is an empty array. The new session starts with a blank, unlocked entry.

There is a second masking behavior: loading that saved file in the same session leaves the taken peak visible because restore replaces the dock's entries only when the saved list is nonempty. The UI reports a successful load while the display and the saved model disagree. S06

Required behavior: save the current observation snapshot independently of fitting U. Restore must replace the list even when it is empty. Saving observations should not silently fit or change the operator's orientation belief.

Acceptance check: save and reload before a first fit; then repeat after editing, removing and locking observations following a fit. Include an empty-list restore in the same session and a fresh-process restart.

Evidence: interfaces/repro_interfaces.py; PEAK_TAKEN, PEAKS_SAVED_WITHOUT_CALCULATE, FIRST_PEAK_AFTER_LOAD_EMPTY_SAVED_LIST and FIRST_PEAK_AFTER_RESTART records.

F3. U5 consumer migration is still a completion blocker

Origin: known, planned compatibility window. Priority: complete before integrated TAVI/ISAR sign-off.

Inspected ISAR main is e37aa900b2b80ce0cee8b308e6306e095d783bc7, dated 1 October. Its client forwards scan/validate bodies without api_version. Current TAVI requires api_version=2, so production request construction reaches the current guard as HTTP 400, api_version_required. The consumer handoff already identifies this unfinished work. S16 S19 S20

Adding the version alone would be incomplete. I generated a canonical response with TAVI's own public_values and ScanJob.snapshot serializers and parsed it through the actual ISAR parser. Intended h=2, k=1, l=0, energy transfer=2.5 meV, and a=b=c=4.05 angstrom became:

- Q indices (0,0,0).
- An alignment scan, with energy_transfer_mev classified as an angle in degrees.
- Cooper-Nathans header energy 0 and lattice lengths (1,1,1).

This is silent fallback on renamed fields, not a failure in TAVI's current serializer. S17

The consumer also still predicts the old scan grid: a command spanning 0 to 1 by 0.4 produces [0,0.4,0.8,1.2] in ISAR's prediction, while current TAVI executes [0,0.4,0.8]. The non-overshoot change must reach acquisition planning as well as parsing. S18

Status correction: ISAR's UB geometry was synced through its PR 40 on 1 October, based on TAVI 010cec6d. The older WIP statement that this geometry copy never synced is stale. The angle-contract migration is the outstanding piece. TAS_MCP was not available in the accessible repositories; its U5 status is unverified, not proven absent.

Acceptance check: canonical/versioned writes, result parsing without silent defaults, units and scan-kind recognition, actual versus predicted grids, and a complete live validate-submit-read-parse cycle against pinned TAVI main. Include TAS_MCP's published names once its checkout is available.

Evidence: consumers/reproduce_contract_gaps.py, canonical_tavi_payload.json, consumer-contract.log, ISAR parity results, and the pinned handoff.

F4. Direct crystal-angle edits leave Q/HKL describing the previous motors

Origin: inherited. Priority: repair geometry/readout synchronization.

The A2/A6 editing handlers update the corresponding energy and wavevector, fixed energy where applicable, energy transfer and curvature. They do not recompute public Q/HKL from the complete resulting motor state. S07

I exercised actual editingFinished signals for all four instruments, both fixed-energy policies, and both crystal angles: 16 cases. Every case left the previous public Q unchanged with no stale marker.

Example: default PUMA followed by A2=60 degrees:

Quantity	Displayed value	From the complete current motor state
Ei	7.2678 meV	7.267843775 meV
Ef	14.7 meV	14.699984097 meV
Q	(3.1028,0,0) inverse angstrom	(2.642255112,-0.642682852,0) inverse angstrom
HKL	(2,0,0)	No longer that point

The Q discrepancy is 0.790659751 inverse angstrom, well beyond rounding. Public parameter reads return the stale values.

Important boundary: Take Position captured ki/kf correctly in all 16 cases, to the display precision. I did not find a stale-energy defect corrupting those captured observations. The defect is the inconsistency between readouts and motor geometry.

Required behavior: after direct crystal/energy changes, refresh Q/HKL from the full current motors while retaining the explicitly selected motors. If any state intentionally remains a target rather than a measured coordinate, expose that distinction explicitly.

Evidence: interfaces/repro_kinematics.py and GUI_KINEMATICS log records.

F5. Explicit zero A2/A6 launches are rejected before transmission handling

Origin: introduced in angle U3, commit 9476a037. Priority: supported-path regression.

On PUMA, these launch parameters were accepted at the pre-U3 revision and are rejected at current main:

```
{"A2": 0, "scan_command1": "A3 0 0 1"}
{"A6": 0, "scan_command1": "A3 0 0 1"}
```

The first now reports Ei=0.0, Ef=14.7; the second Ei=0.0, Ef=0.0; both fail the positive-energy guard. The newly added angle inversion treats angle2k(0)'s no-selection sentinel as a real zero-energy beam. S08

The established contract says a crystal at exact zero take-off transmits and selects no energy. Its selected energy is absent, not zero or a substituted fixed-mode value. The downstream runtime already supports this. A scan command whose angle itself reaches zero can still emit the appropriate transmission point. Mechanical travel limits and the analytic engine's transmission refusal remain applicable. S09 S26

Required behavior: preserve explicit zero motors and independently absent selected energies, retaining legitimate source settings. Also keep a transmitting direct-motor launch out of the later automatic stage-from-Q solve. Merely deleting the positivity check is insufficient.

Acceptance check: compare held-zero launches and scanned-through-zero points in Ki-fixed and Kf-fixed modes, with per-instrument travel and engine policy. Verify null energy metadata and unchanged raw zero motors.

Evidence: probe_zero_launch_history.py, zero-pre-u3.json, zero-current.json, and runtime-probe.json.

F6. The manual tilted-reflection acceptance tool tests the wrong physical crystal

Origin: introduced during UB truth separation, commit 33a4a753. Priority: repair the verification gate.

The existing manual tool describes a crystal rotated 45 degrees and tilted 4 degrees, with operator belief equal to truth. It passes that tilted mount as belief while creating a default state whose U_true remains identity. scan_config copies that identity physical truth. The correct production change separated truth from belief; this caller was not migrated. S10

At the tool's predicted center, the unmodified setup corresponds to actual TAVI HKL approximately (1.41076861,-1.41076861,0.13951295), not (2,0,0). Its intended (200) lab-Q closure error is 2.3817588002 inverse angstrom, an angular miss of 45.1394 degrees. Setting base.U_true to the intended mount before scan_config reduces closure error to 6.94e-16.

The tool also prints obsolete A1/A2/A4 meanings for the monochromator/sample/analyzer take-offs. These must become A2/A4/A6. S10

Required behavior: make the controlled fixture's truth and belief equal explicitly, update its printed angle names, then run its stated acceptance criteria. A correct unit test of the solver cannot repair an incorrect manual fixture.

The audit ran a separately corrected physical control through production snapshot/emission: tilted IN8 gave 44 detector events with solved arcs and zero when those arcs were zeroed, at one million histories per point. This supports the production tilt path; it does not validate the unchanged manual tool or its full fitted-center criterion.

Evidence: ub/check_tilted_gate_truth.py, tilted_gate_truth_result.json, mcstas-runtime/smoke_tilt.py and tilt_results.json.

F7. Negative slit gaps are accepted as feasible and sent to McStas

Origin: validation gap in the slit scan surface introduced by U3. Priority: physical input-domain validation.

The command pre_sample_hgap -1 -1 1 is accepted on all four instruments. The planner reports a feasible point and the runtime emits a negative full gap of -0.001 metres. The hard constraints check binding availability and curvature domains but do not enforce the slit gap's physical domain. S11 S12

A real PUMA McStas run over -1, 0 and +1 mm accepted the negative gap without a negative-width warning. McStas 3.4's Slit component then obtains reversed xmin/xmax bounds, causing the intersection test to absorb every ray. Zero width receives its separate closed-slit warning. Thus a scan can proceed with an impossible negative aperture and produce a misleading closed-beam result.

Required behavior: require finite, nonnegative effective gap values, including after relative offsets are added to frozen bases. Preserve zero as a legitimate closed slit. The same check should cover directly supplied base gaps and planned points.

Acceptance check: relative and absolute scans crossing zero, zero-width behavior, and correct per-instrument mm-to-m emission for positive gaps.

Evidence: runtime-probe.json, mcstas-runtime/slit_results.json and the PUMA slit logs.

F8. Scan coordinates are rounded into repeated positions

Origin: inherited and still present in the new planner. Priority: scan correctness.

The execution grid is rounded to three decimals independently of the quantity's units. The current planner accepts h 1 1.001 0.0001 as eleven points, but six execute at h=1.000 and five at h=1.001. The equivalent A3 scan from 0 to 0.001 by 0.0001 also executes only two distinct positions. S13

The new finite-value, step-sign, point-count and non-overshoot guards are useful, but they do not remove this quantization. A normal UI preference for three displayed decimals should not silently redefine numerical scan positions.

Required behavior: retain full calculation precision through expansion and execution; round only for display. If a physical resolution must be enforced, declare it per quantity and refuse unsupported steps with an explicit message. Consider folder naming and coordinate serialization together when removing rounding.

Acceptance check: compare requested, validated and executed coordinates for small steps, non-dividing endpoints, descending runs and relative scans. Check distinct point identity and output naming without a full simulation.

Evidence: probe_fine_steps.py and fine-steps.json. The pre-UB parser already contains the three-decimal rounding.

3. A retained scientific limitation that wider UB use makes important

F9. Correct centers do not yet imply a correctly oriented resolution ellipsoid

Origin: explicitly documented, pre-existing approximation. Disposition: scientific scope decision or follow-up implementation.

The new true-HKL center transform is correct. However, the deterministic engine still converts reflection offsets by a scalar 2 pi/a and contracts them directly with a resolution matrix whose directions are Q_parallel, Q_perpendicular and Q_vertical. Phonon gradients similarly receive only the cubic scalar conversion. The per-point UB/stage mapping of offsets and gradients is absent. S14 S15

A symmetry-equivalent remount demonstrates the consequence within the model. Using Al_bragg and the PUMA Cooper-Nathans golden geometry, compare an elastic radial scan through (200) along H against a remount making a radial scan through (020) along K physically identical. The mounted Q vectors agree to about 2e-16 inverse angstrom, and both settings solve to the same A3/A4 with zero arcs.

Radial offset, r.l.u.	H-radial mean counts	Equivalent K-radial mean counts
0	100.00000	100.00000
0.005	90.03329	86.90415
0.010	65.87509	57.19705
0.020	19.21594	10.94157

The last K-based prediction is 43.1 percent lower solely because a radial offset is assigned to the perpendicular resolution direction. These are deterministic model outputs at ten million histories, not measured experimental counts or a new calibration claim.

This limitation was present before the UB work and is acknowledged in the analytic-engine documentation. It should not be presented as a newly introduced undisclosed regression. It does mean that arbitrary UB support in targeting and fitting is broader than the currently justified orientation support of quantitative analytic convolution.

Required behavior for general mounts: provide a per-point transformation from reciprocal coordinates to the instrument resolution basis. Transform reflection offsets with that map and phonon gradients with the corresponding inverse transpose. Include the true B metric, not only a cubic scale.

Acceptance check: invariant widths and intensities for symmetry-equivalent remounts, tilted mounts and noncubic metrics, across every convolution implementation. Until then, keep the applicable approximation clear in scientific use.

Evidence: ub/check_resolution_frame.py and resolution_frame_result.json.

4. Verification results and their limits

Independent numerical checks

These checks used deterministic random seeds and separate numerical constructions, including a SciPy rotation implementation and a closed-form crossed-arc reachability oracle. They are stronger evidence than merely round-tripping the same production helper.

Check	Coverage	Result
Reciprocal metric	300 random valid nonorthogonal cells	B^T B=(2 pi)^2 G^-1; maximum absolute difference 8.24e-13
Stage reachability	4,000 random Q vectors at two arc travel limits	1,512 accepted, 2,488 refused; zero oracle disagreements
Forward/inverse closure	Accepted random points	Maximum Q errors 4.32e-15 / 4.43e-15
McStas arm reconstruction	Independent stage and true rotations	Maximum matrix-element error 9.99e-16
Two-/multi-peak fitting	200 random orientations, four lattice families, six HKLs, both senses	Maximum U error 1.42e-15 / 5.33e-15
Lattice refinement	All seven crystal systems	Maximum lattice-parameter error 5.68e-14; maximum residual 1.86e-15
Plane locking	400 random attempts, 25 reachable	Maximum levelled-normal error 3.42e-16

The differing numerical units and scales matter: these are maximum absolute discrepancies for each stated check, not a shared physical accuracy specification.

Repository suite: complete run plus focused environment repair

The initial full TAVI run produced **2,356 passed, 67 failed, 32 errors, 193 skipped** in 276.09 seconds. It was run once, with isolated configuration and offscreen Qt, respecting the single-suite rule.

Of the failures/errors, 49 failures and all 32 errors were model-building paths blocked by the absent McStas resources. After extracting official McStas 3.4 into scratch, a focused rerun of the four build-tree modules and plugin bindings passed **all 96 tests**. This resolves that environment-related group without claiming that a second full suite ran.

The remaining TAVI failures are:

- **16 layout parameterizations:** a mixture of actual clipping and overly strict preferred-size expectations. They are not 16 independent defects.
- **2 orientation goldens:** exact list equality differs by only 2.22e-16 and 3.89e-16. Historical and current implementations are bit-identical under the same environment. These failures do not demonstrate a UB regression or an absent TAS_MCP dependency. S22

The 193 skips comprise 187 Windows-specific cases, three unreachable stage fixtures, one absent Pb DFT map, one external documentation checker and one McStas resource-locator case. Windows installation and native rendering therefore remain outside this audit's executed coverage.

ISAR's geometry suite separately produced **69 passed, 1 failed**. The failing stage golden uses a 1e-9-degree threshold on optimizer-selected angles. Across 96 fixture points, the largest recorded-angle

drift was 4.47e-7 degrees; current ISAR and current TAVI returned exactly the same angles on this host, and the worst physical Q closure residual was 3.42e-15 inverse angstrom. This is a brittle numerical golden, not geometry-vendor drift.

Environment: Linux x86_64; Python 3.12.14; NumPy 2.5.4; SciPy 1.18.1; PySide6 6.12.0; pytest 9.1.1; mcstasscript 0.0.95; McStas 3.4. Version and test evidence are retained in the bundle.

Real McStas execution

All four emitted instrument models generated C, compiled with gcc and completed serial 10,000-history runs with detector output and the new physical parameter names. Those first runs had zero detector events and establish compilation/plumbing only.

The subsequent one-million-history controls establish a physical Bragg response:

Control	Predicted/solved setting	Deliberately displaced control
PUMA, sample sense -1, nominal (200)	214 detector events	0 at A3 +2 degrees
IN8, sample sense +1, nominal (200)	44 detector events	0 at A3 +2 degrees
IN8, true tilted mount	44 detector events at solved arcs	0 with both arcs zeroed

The tilted case used the physical truth equal to the described mount, rotated 45 degrees and tilted 4 degrees. Production solve gave A3=-9.30491019, sgl=-2.82727441 and sgu=-2.83072876 degrees at Ei=Ef=14.7 meV. All controls used seed 812 and production snapshot/parameter emission.

These controls support rotation/sign conventions and the truth-stage-engine path. They are not a full rocking-curve fit, a completed UB collection/fit/restart exercise, MPI validation, or a calibrated intensity benchmark.

5. Cleanup and remaining acceptance work

Layout and test cleanup

The new inline crystal readouts can crop significant digits under the tested Linux Fusion/DejaVu Sans 9-point layout. After committing A2=123.4567, the 50-pixel field contains 57 pixels of text; leading digits scroll out of view. A1=61.7283 is similarly cramped. Stored angles remain correct. The inline layout change is commit 6d05635, within the angle programme. S21

The long runtime estimate also clips, and the reflection-table checkbox is marginally too narrow. Those are inherited compact-layout issues. The Valid Commands button, in contrast, renders its text completely: its preferred-size test is too strict. Prefer checks of actual content rectangles or intentional elision with complete-text access over a blanket width-versus-sizeHint assertion.

The orientation compatibility goldens should use a stated tight tolerance and retain physical truth/orthogonality checks. The fingerprint test still sets an obsolete A3 attribute rather than sample_rotation_deg; other changed values make the test pass, so its turntable coverage is weaker than the name suggests. S22 S23

Update stale comments and status records: old take-off numbering in runtime/manual helpers, the correction-record docstring, and the WIP statement that ISAR geometry never synced. The approved deferred internal-key rename should remain a deliberate maintenance item; old internal identifiers alone do not establish a public-contract failure. S20

What I would require for closeout

1. **Trustworthy state:** close F1, F2 and F4, with tests through actual user/API entry points and fresh-session reload.
2. **Correct scan boundaries:** close F5, F7 and F8 using the same expansion and runtime authority used by validation and execution.
3. **Trustworthy acceptance tooling:** repair F6 and the obsolete test fields, then run the manual tilted-reflection gate's full fitted-center, contrast and significance criteria.
4. **Complete consumers:** land U5, verify the real ISAR request/parse cycle and grid contract, and inspect TAS_MCP's actual published interface.
5. **State the scientific scope:** either implement F9's resolution-frame mapping or keep quantitative general-mount analytic claims explicitly limited.
6. **Finish operator acceptance:** exercise peak capture, fitting, grading, plane locking, save/restart and the new field marks on the intended desktop platform.

Limits that remain material

The positive core result does not establish arbitrary noncubic physical sample transport. The shipped Single_crystal sample is cubic FCC Al. McStas 3.4's default component basis differs from TAVI's convention; cubic symmetry makes the elastic reciprocal set invariant under that permutation. This retained boundary deserves an explicit component contract before extending general noncubic transport claims. The Python nonorthogonal geometry checks remain valid.

The complete Windows-specific suite, native desktop layout, missing Pb DFT asset path, external documentation checker, MPI runs, and TAS_MCP checkout were not verified here. The real transport controls exercised production models and snapshots with direct serial mcrun invocation; they did not exercise every GUI queue or first-point wrapper path.

This audit supplies a concrete closeout list and reproducible evidence. The examined mathematical changes are supported by the checks; declaring both programmes entirely complete would require resolving or explicitly accepting the items above.

Evidence navigation

The companion archive contains:

- **README.txt:** pinned revisions, setup and rerun guidance.
- **interfaces/**: actual controller, persistence, editing-signal and layout probes, logs and screenshots.
- **ub/**: independent metrics/reachability/fitting checks, analytic-frame comparison, historical golden comparison and manual-gate truth reproduction.
- **consumers/**: canonical TAVI response, ISAR contract reproduction and geometry-parity evidence.
- **runtime/**: scan-grid, transmission and slit preflight probes, including the pre-U3 comparison.
- **mcstas-runtime/**: exact smoke/control scripts, package provenance, parameter snapshots, emitted .instr files, logs and detector outputs.
- **tests/**: the full TAVI run, focused McStas build rerun and ISAR geometry verification summary.

Only audit evidence and reproducible scripts are included. The production repositories and downloaded McStas distribution remain identified by revision/provenance rather than duplicated as report artifacts.

Source index

All TAVI links below are pinned to the audited main revision. ISAR links are pinned to its separately inspected main revision. Historical comparisons are identified by full commit above and reproduced in the archive.

S01 - Angle, frame, truth and belief conventions. TAVI / docs/INSTRUMENT_LAYOUT.md#L25-L164

S02 - Declared PATCH dependency order. TAVI / TAVI_PySide6.py#L7813-L7826

S03 - PATCH setters and callbacks. TAVI / TAVI_PySide6.py#L8066-L8094

S04 - Save serializes the UB model. TAVI / TAVI_PySide6.py#L5969-L5992

S05 - UB model observation serialization. TAVI / tavi/ub_matrix.py#L746-L760

S06 - Empty observation lists are not restored. TAVI / TAVI_PySide6.py#L6475-L6488

S07 - Direct crystal-angle editing handlers. TAVI / TAVI_PySide6.py#L3857-L3932

S08 - Explicit-angle API launch derivation. TAVI / TAVI_PySide6.py#L2975-L3072

S09 - Approved direct-transmission semantics. TAVI / docs/CONFIGURABLE_INSTRUMENTS.md#L1885-L1902

S10 - Manual tilt-gate truth omission and old labels. TAVI / tools/check_tilted_reflection.py#L153-L176

S11 - Scan hard constraints and curvature-domain check. TAVI / instruments/rules.py#L371-L407

S12 - Slit values emitted to runtime parameters. TAVI / instruments/tas_runtime.py#L1359-L1368

S13 - Three-decimal scan-coordinate rounding. TAVI / tavi/utilities.py#L96-L141

S14 - Analytic convolution frame. TAVI / tavi/deterministic_engine.py#L598-L624

S15 - Documented general-UB approximation. TAVI / docs/ANALYTIC_ENGINE.md#L189-L194

S16 - ISAR unversioned writes. ISAR / isar/drive/instrument/tavi_client.py#L117-L134

S17 - ISAR legacy quantity parsing. ISAR / isar/parsers/tavi.py#L212-L401

S18 - ISAR old scan-grid rule. ISAR / isar/drive/acquisition/grid_adapters.py#L89-L123

S19 - U5 consumer migration handoff. TAVI / docs/HANDOFF-scan-command-consumers.md

S20 - Open acceptance work and consumer status. TAVI / WIP.md#L163-L241

S21 - Inline crystal-angle/readout sizing. TAVI / gui/docks/instrument_dock.py#L72-L97

S22 - Strict orientation golden comparison. TAVI / tests/test_orientation.py#L303-L336

S23 - Obsolete A3 field in fingerprint test. TAVI / tests/test_orientation.py#L1018-L1030

S24 - Stage composition, peak inverses and physical arm transform. TAVI / tavi/orientation.py

S25 - UB fitting, refinement and grading. TAVI / tavi/ub_matrix.py

S26 - Independent selected energies and transmission. TAVI / instruments/tas_runtime.py#L767-L796